Resultater fra QH4 for Hilde Celine Løvland

Resultat for dette forsøket: 0 av 11
Innlevert 24. sep. i 16.28
Dette forsøket tok under 1 minutt

Ubesvart

Spørsmål 1
0 / 5 poeng

Hydraulic system . Consider a fixed displacement pump connected to a base unit, see Figure Q4.7 . The base unit is a 4/3-way proportional DCV and a retracting cylinder, see Figure Q4.5 . Pump displacement (D) = 56 cm^3/rev . Pump speed (n) = 1030 rev/min . Pump volumetric efficiency (eta_vP) = 0.92 . Main crack pressure (p_crM) = 190 bar . The 4/3-way proportional directional control valve has identical discharge coefficients: C_dPA = C_dBT = C_dPB = C_dAT = C_d . The 4/3-way proportional directional control valve has symmetrical discharge areas: A_dMax,PA = A_dMax,PB and A_dMax,BT = A_dMax,AT . Liquid density (rho) = 875 kg/m^3 . Discharge coefficient (C_d) = 0.70 . Maximum discharge area (A_dMax,PB) = 39 mm^2 . Maximum discharge area (A_dMax,AT) = 45 mm^2 . Max valve spool travel (x_Max) = 10.0 mm . Valve spool travel (x) = -8.0 mm . The cylinder is retracting . The cylinder is subjected to a resistive (positive) load . Cylinder diameter (d) = 25 mm . Cylinder rod diameter (dR) = 16 mm . Cylinder hydromechanical efficiency (eta_hmC) = 0.92 . Cylinder load force (F_L) = 4.93 kN . What is the pressure (p_B) in bar?

188,9037 (med marginen: 1,8891)

Ubesvart

Spørsmål 2
0 / 1 poeng

Hydro-mechanical eigenfrequency . Consider a motor connected to a servo valve, see Figure Q4.10 . The two hydraulic lines are identical, i.e., V_L1 = V_L2 and V_L = V_L1 + V_L2 . Bulk modulus (beta) = 1053 MPa . Motor displacement (D) = 250 cm^3/rev . Total line volume (V_L) = 0.63 l . Mass moment of inertia (J) = 5.63 kg*m^2 . What is the natural eigenfrequency (w_n) in rad/s?

36,6867 (med marginen: 0,3669)

Ubesvart

Spørsmål 3
0 / 2 poeng

Minimum rated flow for servo valve . Consider a motor connected to a servo valve, see Figure Q4.12 . Motor displacement (D) = 160 cm^3/rev . Mass moment of inertia (J) = 2.70 kg*m^2 . Supply pressure (p_s) = 250 bar . Payload total rotation (Delta-theta) = 73.1 rad . Total task time (Delta-t) = 2.525 s . Ramp time (t_R) = 0.569 s . External moment (M_1) = 49 . External moment (M_2) = -347 . External moment (M_3) = 334 . Consider a specific time during the task (t) = 1.810 s . What is the no-load flow (Q_NL) at the considered time (t) in l/min?

45,9374 (med marginen: 0,4594)

Ubesvart

Spørsmål 4

0 / 3 poeng

No-load flow in servo systems . Consider a motor connected to a servo valve, see Figure Q4.12 . Motor displacement (D) = 355 cm^3/rev . Mass moment of inertia (J) = 9.63 kg*m^2 . Supply pressure (p_s) = 250 bar . Valve rated pressure (p_r) = 70 bar . Payload total rotation (Delta-theta) = 81.8 rad . Total task time (Delta-t) = 6.149 s . Ramp time (t_R) = 1.703 s . External moment (M_1) = 453 . External moment (M_2) = -217 . External moment (M_3) = -19 . What is the minimum required valve flow (Q_r,min) in l/min?

42,4089 (med marginen: 0,4241)

Quizresultat: 0 av 11